Southern New Hampshire University

CS 505 Module Eight Activity

Part One: Functions and Property Analysis

For Part One, solve the problems provided, which pertain to functions and properties.

1. Define a function *f* that is both one-to-one (injective) and onto (surjective) for the provided function, including an explanation of the definition:
   Let f: $A \rightarrow B$ be a function, where $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$.

   Answer:
   $f(A) = B$:
   $f(1) = a$
   $f(2) = b$
   $f(3) = c$
   or
   $f = \{(1, a), (2, b), (3, c)\}$

   This function is both injective and surjective. It is injective because every element of A maps to a unique elen
   no two elements of A map to the same element in B. In addition, it is surjective because every value in B is mapped to by a value in A. Therefore, the function is bijective.

2. Define a function *f* that is one-to-one (injective) but not onto (surjective) for the provided function, including an explanation of the definition:
   Let f: $A \rightarrow B$ be a function, where $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$.

   Answer:
   Assuming this question is free of errors, it is impossible for a function to exist with these sets to be injective but not surjective. Given the two sets, injectivity means that A maps each element to a unique element in B. When each element in A is mapped to a unique element in B, it is automatically surjective, since B contains at least one element mapped from A.

   A modification to this which would allow injectivity but not surjectivity would be adding ‘d’ to the set ‘B’. This would allow A to map to unique elements in B but would avoid surjectivity because not all elements in B would be mapped by A.

   If the modification existed, a function that would fit this is:
   Let f: $A \rightarrow B$ be a function, where $A = \{1, 2, 3\}$ and $B = \{a, b, c, d\}$.
   $f(A) = B$:
   $f(1) = a$
   $f(2) = b$
   $f(3) = c$
   or
   $f = \{(1, a), (2, b), (3, c)\}$

3. Explain whether R and S are reflexive, symmetric, or transitive for the information provided:
   Let R and S be relations on the set $A = \{1, 2, 3\}$ defined as
   $R = \{(1, 3), (1, 2), (2, 2)\}$ $S = \{(1, 2), (2, 3), (3, 1)\}$.

   Answer:
   A relation is **reflexive** if every element in a set is related to itself: $(a, a) \in R$.
   A relation is **symmetric** if whenever $(a, b) \in R$, then $(b, a) \in R$.
   A relation is **transitive** if whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$.

$R = \{(1,3), (1,2), (2,2)\}$
   R is **not** reflexive because (1,1) and (3,3) are missing.
   R is **not** symmetric because (1,3) is in R, but (3,1) is not.
   R is transitive because all required transitive pairs are present.

$S = \{(1,2), (2,3), (3,1)\}$
   S is **not** reflexive because (1,1), (2,2), and (3,3) are missing.
   S is **not** symmetric because (1,2) is in S, but (2,1) is not.
   S is **not** transitive because (1,2) and (2,3) are in S, but (1,3) is not.

4. Explain whether R in the information provided is reflexive, symmetric, and/or transitive:
   Let $A = \{1, 2, 3\}$ and R be a relation on A defined as
   $R = \{(1, 1), (1, 3), (2, 1), (2, 3), (3, 3)\}$.

   Answer:
   A relation is **reflexive** if every element in a set is related to itself: $(a, a) \in R$.
   A relation is **symmetric** if whenever $(a, b) \in R$, then $(b, a) \in R$.
   A relation is **transitive** if whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$.

$R = \{(1,1), (1,3), (2,1), (2,3), (3,3)\}$
   R is **not** reflexive because $(2,2)$ is missing.
   R is **not** symmetric because the following are missing: $(3,1), (1,2), (3,2)$
   R is transitive because all required transitive pairs are present

5. Explain whether a relation is reflexive for the information provided:
   Consider the set of all people in a town, denoted as P, and define a relation R where $(a, b) \in R$ means "person a is a sibling of person b."
   Answer:
   Let P be the set of all people in the town, and let R be the relation where (a,b)∈R means "person a is a sibling of person b." The relation is **not reflexive** because a person is not considered to be their own sibling. Therefore, $(a, a) \notin R$ for every person $a \in P$.

6. Explain whether a relation is symmetric for the information provided:
   Consider the set of all people in a town, denoted as P, and define a relation R where $(a, b) \in R$ means "person a is a sibling of person b."

   Answer:
   Let P be the set of all people in the town, and let R be the relation where $(a, b) \in R$ means "person a is a sibling of person b." The relation is symmetric because if person a is a sibling of person b, then person b is also a sibling of person a. Therefore, whenever $(a, b) \in R, (b, a) \in R$

7. Explain whether a relation is transitive for the information provided:
   Consider the set of all people in a town, denoted as P, and define a relation R where $(a, b) \in R$ means "person a is a sibling of person b."

   Answer:
   Let P be the set of all people in the town, and let R be the relation where $(a, b) \in R$ means "person a is a sibling of person b." The relation is **not transitive** because it is possible for person a to be a sibling of person b, and for person b to be a sibling of person c, while person a is not a sibling of person c. For example, this can occur in families with half-siblings. Therefore, the relation does not satisfy the transitive property.

Part Two: Equivalence Relations

For Part Two, solve the problems provided, which pertain to equivalence.

8. Prove that R is an equivalence relation for the information provided:
   Let R be a relation on the set of integers Z defined as
   $R = \{(x, y) \mid x \equiv y \text{ (mod 6)}\}$.

   Answer:
   Reflexive:
   For any integer x, $x - x = 0$. Since 0 is divisible by 6, $x \equiv x\ (mod\ 6)$.
   So, $(x, x) \in R$; therefore, R is reflexive.

   Symmetric:
   Assume $x \equiv y\ (mod\ 6)$, meaning that 6 divides $x - y$. This means that $x - y = 6k$ for some integer k.
   Multiplying both sides by -1: $y - x = -6k$. Since $-6k$ is divisible by 6, this means $y \equiv x\ (mod\ 6)$. Therefore, R is symmetric.

   Transitive:
   Assume $x \equiv y\ (mod\ 6)$ and $y \equiv z\ (mod\ 6)$. This means: $x - y = 6m$ and $y - z = 6n$, respectively, for some integers m and n.
   Adding these together: $(x - y) + (y - z) = 6m + 6n$
   Simplified: $x - z = 6(m + n)$
   Since $m + n$ is an integer, $x - z$ is divisible by 6.
   Therefore, R is transitive.

   Since R is reflexive, symmetric, and transitive, R is an equivalence relation.

9. Describe the equivalence classes of R for the information provided:
   Let R be a relation on the set of integers Z defined as
   $R = \{(x, y) \mid x \equiv y \pmod 6\}$.

   Answer:
   Given $R = \{(x, y)\} \mid x \equiv y\ (mod\ 6)$:
   This can be broken down into 6 equivalence classes based on modulo 6.
   The equivalence classes are $[0], [1], [2]. [3]. [4]. [5]$.
   This is because every integer has 1 of 6 possible remainders when divided by 6: 0, 1, 2, 3, 4, or 5.
   $[0] = \{\dots, -12, -6, 0, 6, 12, \dots\}$
   $[1] = \{\dots, -11, -5, 1, 7, \dots\}$
   $[2] = \{\dots - 10., -4, 2, 8, \dots\}$
   $[3] = \{\dots, -9, -3, 3, 9, \dots\}$
   $[4] = \{\dots, -8, -2, 4, 10, \dots\}$
   $[5] = \{\dots - 7, -1, 5, 11, \dots\}$

Part Three: Composition of Relations and Function Inverse
For Part Three, solve the problems provided, which pertain to relations and function inverse.

10. Describe the compositions $R \circ S$ and $S \circ R$ for the information provided:
    Let R and S be relations on the set $A = \{1, 2, 3\}$ defined as
    $R = \{(1, 3), (1, 2), (2, 2)\}$ $S = \{(1, 2), (2, 3), (3, 1)\}$.

    Answer:
    $R \circ S = \{(x, z) | (x, y) \in S\ and\ (y, z) \in R$
    $(1, 2) \in S\ and\ (2, 2) \in R$: Therefore, $(1,2) \in R \circ S$
    $(2, 3) \in S$ but there is nothing in R beginning with 3.
    $(3,1) \in S\ and\ (1,3), (1,2) \in R$: Therefore, $(3,3)\ and\ (3,2) \in R \circ S$
    Therefore: $\boldsymbol{R \circ S = \{(1, 2), (3, 3), (3, 2)\}}$

    $S \circ R = \{(x, z) | (x, y) \in R\ and\ (y, z) \in S\}$
    $(1,3) \in R\ and\ (3,1) \in S$: Therefore, $(1,1) \in S \circ R$
    $(1,2) \in R\ and\ (2,3) \in S$: Therefore, $(1,3) \in S \circ R$
    $(2,2) \in R\ and\ (2,3) \in S$: Therefore, $(2,3) \in S \circ R$
    $\boldsymbol{Therefore: S \circ R = \{(1, 1), (1, 3), (2, 3)\}}$

11. Prove that the inverse function f^{-1}: $B \rightarrow A$ is also bijective, assuming that f: $A \rightarrow B$ is a bijective function.

    Answer:
    Since: $f: A \rightarrow B$ is bijective, it is both one-to-one and onto. Therefore, every element of $b \in B$ corresponds to exactly one element $a \in A$, so the inverse function $f^{-1}: B \rightarrow A$ exists.

    $f^{-1}: B \rightarrow A$ is One-to-One:
    Assume: $f^{-1}(b_1) = f^{-1}(b_2)$
    This means both b_1 and b_2 map back to the same element in A.
    Let: $f^{-1}(b_1) = a \; and \; f^{-1}(b_2) = a$
    Then: $f(a) = b_1 \; and \; f(a) = b_2$
    Therefore: $b_1 = b_2$ so f^{-1} is one-to-one.

    $f^{-1}: B \rightarrow A$ is Onto:
    To show $f^1$is onto, choose any $a \in A$.
    Since f maps from $A \; to \; B$, there is some $b \in B$ such that: $f(a) = b$
    Then by definition of inverse: $f^{-1}(b) = a$
    So every element of A is reached by f^{-1}. Therefore, f^{-1} is onto.

    Since f^{-1} is both one-to-one and onto, $f^{-1}: B \rightarrow A$ is bijective.

Part Four: Recursion and Induction

For Part Four, solve the problems provided, which pertain to recursion and induction.

12. Prove the correctness of the recursive factorial function using mathematical induction.

    Answer:
    Let the recursive factorial function be defined as:
    $$factorial(n) = \begin{cases} 1, & n = 0 \\ n \cdot factorial(n-1), & n > 0 \end{cases}$$
    Base Case:
    When $n = 0$, $factorial(0) = 1$
    By definition, $0! = 1$, so the function is correct for $n = 0$.

    Inductive Hypothesis:
    Assume the function is correct for some integer $k \geq 0$:
    $factorial(k) = k!$

    Inductive Step:
    Recursive definition: $factorial(k+1) = (k+1) \cdot factorial(k)$
    By the inductive hypothesis: $factorial(k) = k!$
    So: $factorial(k+1) = (k+1) \cdot k!$
    By the definition of factorial: $(k+1)! = (k+1) \cdot k!$
    Therefore: $factorial(k+1) = (k+1)!$

    Since the base case is true and the inductive step is true, by mathematical induction, the recursive factorial function correctly computes $n!$ for all integers $n \geq 0$.

13. Prove the correctness of the recursive Fibonacci function using mathematical induction.

Answer:
Let the recursive Fibonacci function be defined as:

$$f(n) = \begin{cases} 0, & n = 0 \\ 1, & n = 1 \\ f(n-1) + f(n-2), & n \geq 2 \end{cases}$$

Base Case:
$f(0) = 0$ and $f(1) = 1$, which match the definition of the Fibonacci sequence.

Inductive Hypothesis:
$f(k)$ and $f(k-1)$ are equal to the kth and $(k-1)$th Fibonacci numbers, respectively.

Inductive Step:
Recursive definition: $f(k+1) = f(k) + f(k-1)$
By the inductive hypothesis, both $f(k)$ and $f(k-1)$ are correct Fibonacci numbers, so their sum is exactly the next Fibonacci number.
Therefore: $f(k+1)$ is equal to the $(k+1)th$ Fibonacci number and is computed correctly.

Since the base cases are correct and the inductive step is true, by mathematical induction, the recursive Fibonacci function correctly computes the Fibonacci sequence for all $n \geq 0$.

14. Describe the time complexity of the recursive function provided using recurrence relations:

```
def recursive_sum(arr, n):
    if n == 0:
        return 0
    else:
        return arr[n - 1] + recursive_sum(arr, n - 1)
```

Answer:
The recurrence relation for the function is: $T(n) = T(n-1) + 1$
Where:

$T(0) = 1$ (base case)
The $+1$ represents the constant-time work performed during each recursive call

Expand:
$T(n) = T(n-1) + 1$
$T(n) = T(n-2) + 2$
$T(n) = T(0) + n$
Since $T(0)$ is a constant: $T(n) = n + 1$
Therefore, the time complexity is: $O(n)$ because the function makes one recursive call for each element in the array.

15. Describe the space complexity of the recursive function provided using recurrence relations:

```
def recursive_sum(arr, n):
    if n == 0:
        return 0
    else:
        return arr[n - 1] + recursive_sum(arr, n - 1)
```

Answer:
The recurrence relation for space usage is: $S(n) = S(n-1) + 1$
Where: $S(0) = 1$ (base case) and the $+1$ represents one additional stack frame created by each recursive call.

Expand:
$S(n) = S(n-1) + 1$
$S(n) = S(n-2) + 2$
$S(n) = S(0) + n$
Since $S(0)$ is constant: $S(n) = n + 1$
Therefore, the space complexity is $O(n)$ because the recursion stack grows by one frame for each recursive call until the base case is reached.

16. Explain why a recursive approach is inefficient compared to an iterative solution for the recursive function provided:

```
def recursive_sum(arr, n):
    if n == 0:
        return 0
    else:
        return arr[n − 1] + recursive_sum(arr, n − 1)
```

Answer:
Although both recursive and iterative solutions have O(n) time complexity, the recursive approach is less efficient because it requires additional memory for the call stack. Each recursive call creates a new stack frame that stores parameters, return addresses, and local variables.

The recursive version has a space complexity of O(n), whereas an iterative solution requires only a single running total and a loop counter, resulting in a space complexity of O(1).

Additionally, recursive calls introduce function call overhead, making the recursive solution slower in practice. For very large arrays, deep recursion may also cause a stack overflow, whereas an iterative solution can handle much larger inputs without this risk.

Therefore, an iterative solution is generally more memory-efficient and performs better than the recursive approach for this problem.